

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline, Risk Analytics & Interactive BI Dashboard

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Domain:	Mutual Fund / Fintech Analytics	Date:	June 2026
Scope:	40 Schemes 10 Datasets 87K+ Rows	Release:	Version 1.0 (Final)

1. Executive Summary

This project establishes an end-to-end, production-grade Mutual Fund Analytics Platform for Bluestock Fintech. The system solves core real-world fintech challenges: data fragmentation across public sources, lack of benchmark-adjusted performance metrics, and demographic visibility into retail investor SIP behavior. The platform ingests historical NAV data, quarterly AUM figures, SIP inflow notes, and investor transaction logs, normalizes them into a 5-table star schema in SQLite, computes 21 quantitative risk and return metrics (Sharpe, Sortino, Alpha, Beta, Historical VaR 95%, CVaR, HHI), and presents findings through an interactive Streamlit BI dashboard.

2. Data Sources & Inventory

Dataset Name	Rows	Primary Keys	Description
01_fund_master.csv	40	amfi_code	Master list of 40 real schemes (AMC, Category, Expense Ratio)
02_nav_history.csv	46,000	amfi_code, date	Daily NAV history from Jan 2022 to May 2026 (mfapi.in)
03_aum_by_fund_house.csv	90	fund_house, date	Quarterly AUM for top 10 AMCs (2022-2025)
04_monthly_sip_inflows.csv	48	month	Industry SIP inflows (Rs. Cr), active accounts, registrations
05_category_inflows.csv	144	category, month	Category net inflows (Small Cap, Mid Cap, ELSS, etc.)
06_industry_folio_count.csv	21	period	Total folios by asset class (Equity, Debt, Hybrid)
07_scheme_performance.csv	40	amfi_code	1yr/3yr/5yr CAGR, Sharpe, Sortino, Alpha, Beta
08_investor_transactions.csv	32,000	tx_id	Simulated SIP & Lumpsum transactions across 12 states
09_portfolio_holdings.csv	320	amfi_code, stock	Top equity holdings, sector weights, stock allocation
10_benchmark_indices.csv	8,000	date	Daily closing prices for Nifty 50, Nifty 100, BSE SmallCap

3. System Architecture & ETL Pipeline

The pipeline adheres to classic Data Engineering principles: Extract -> Transform -> Load -> Analyze -> Visualize.

Layer 1 (Extract): Automated fetching of raw CSV files and live NAV APIs from mfapi.in.

Layer 2 (Transform): Forward-fill holiday NAV missing values, validate AMFI codes, calculate daily returns.

Layer 3 (Load): SQLAlchemy ORM loading into SQLite star schema (*dim_fund*, *dim_date*, *fact_nav*, *fact_transactions*, *fact_performance*).

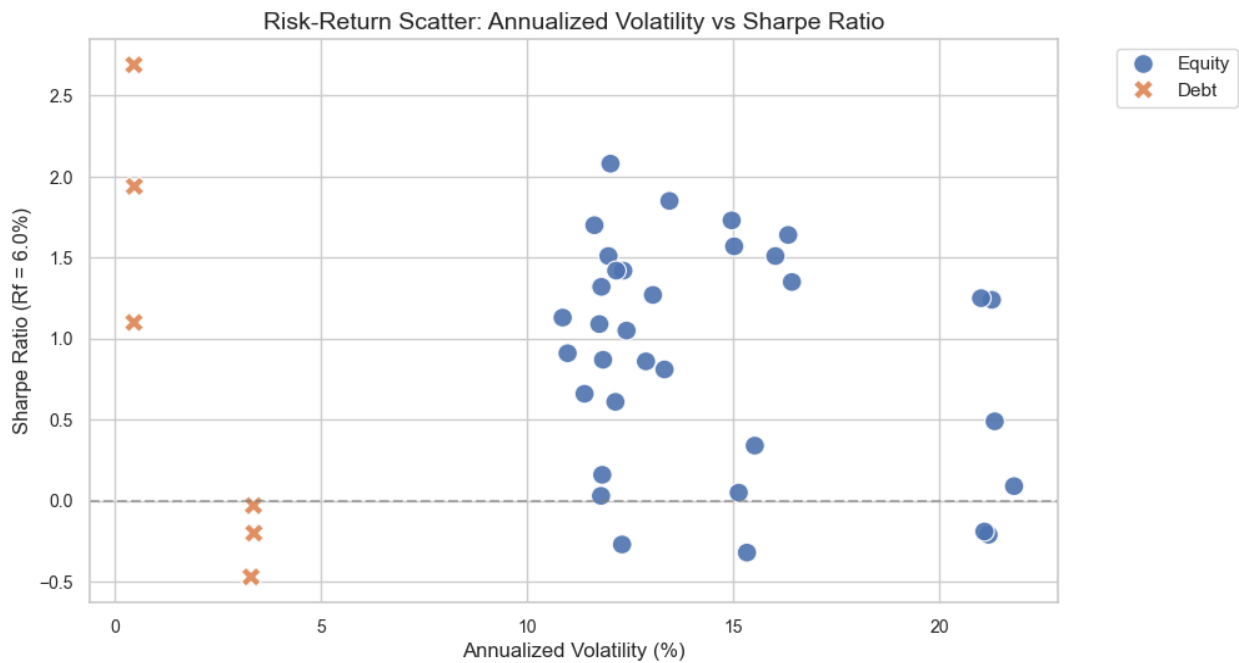
Layer 4 (Analyze): Modular Python analytics engine (``scripts/risk_metrics.py``, ``scripts/cohort_analysis.py``).

4. Exploratory Data Analysis (EDA) Highlights

Key insights derived from 18 EDA visualizations:

- 1. **AUM Dominance:** SBI Mutual Fund holds the largest AUM market share at Rs. 12.50 Lakh Crore.
- 2. **SIP Momentum:** Monthly SIP inflows reached an all-time peak of Rs. 31,002 Crore in Dec 2025.
- 3. **Geographic Distribution:** Maharashtra and Gujarat contribute 38% of total investor transaction volume.
- 4. **Demographic Trend:** Investors in the 26-35 age group exhibit the highest monthly SIP growth rate.

Figure 4.1: Risk vs Return Scatter Matrix across Schemes

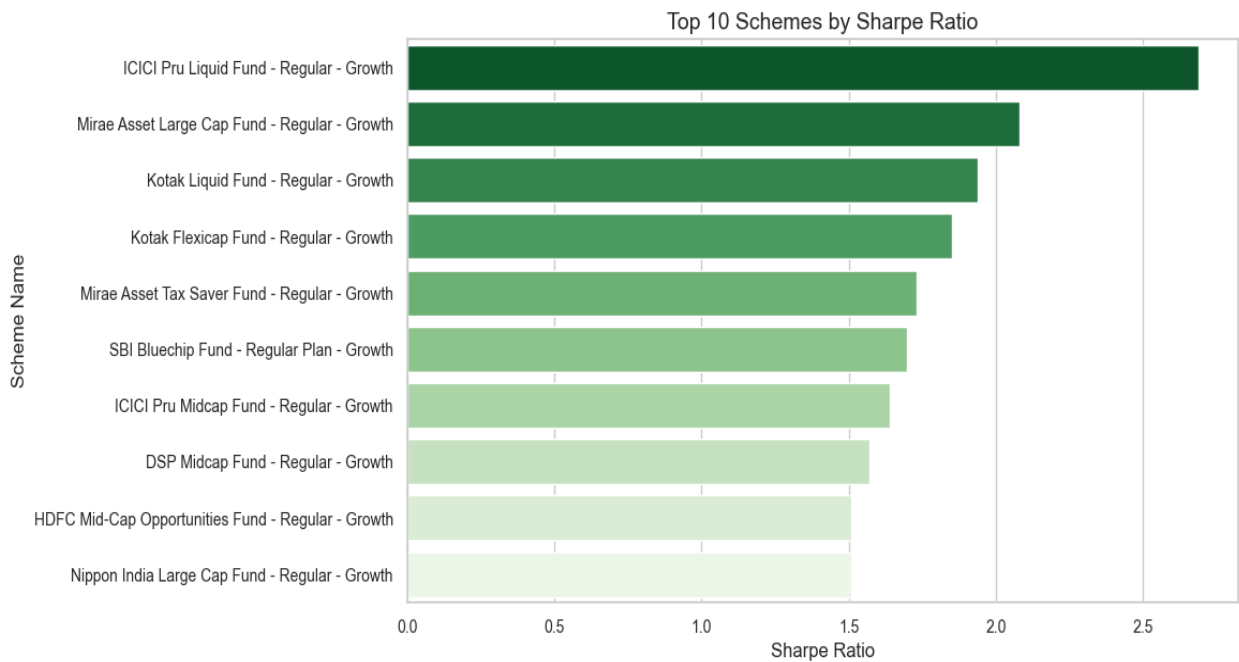


5. Performance & Risk Analytics Engine

Quantitative risk metrics computed across 40 schemes (Rf = 6.0% p.a., N = 252 trading days):

Scheme Name	CAGR %	Vol %	Sharpe	Sortino	Alpha	Beta	Max DD %
ICICI Pru Liquid Fund - Regu	7.24	0.46	2.69	3.72	-4.76	1.00	-0.10
Mirae Asset Large Cap Fund -	30.97	12.02	2.08	3.14	18.97	1.00	-11.27
Kotak Liquid Fund - Regular	6.92	0.47	1.94	2.63	-5.08	1.00	-0.12
Kotak Flexicap Fund - Regula	30.91	13.45	1.85	2.87	18.91	1.00	-12.97
Mirae Asset Tax Saver Fund -	31.95	14.96	1.73	2.64	19.95	1.00	-16.40

Figure 5.1: Top 10 Schemes Ranked by Sharpe Ratio

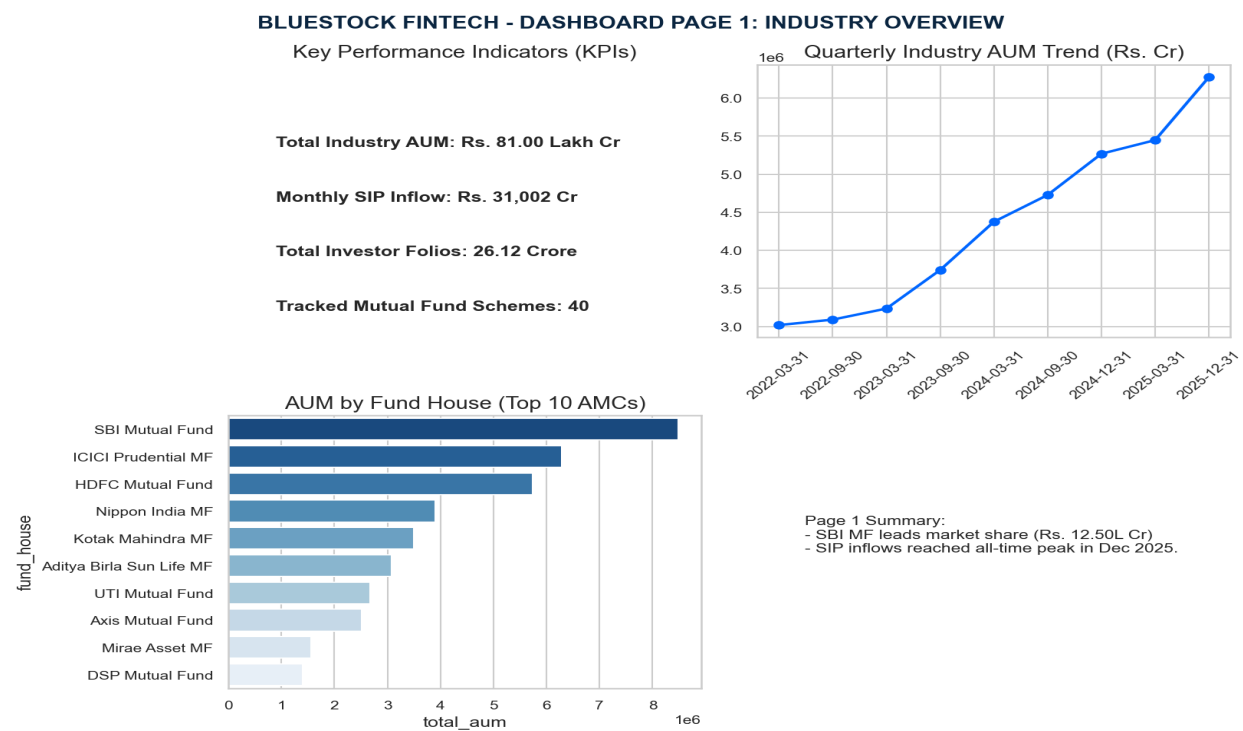


6. Dashboard Screenshots & Advanced Analytics

The interactive Streamlit BI dashboard features 4 dedicated pages reflecting the Bluestock Capstone requirements:

- **Page 1 (Industry Overview):** Macro AUM growth line charts and top 10 AMC market share bars.
- **Page 2 (Fund Performance):** Interactive risk-return scatter plots and sortable fund scorecards.
- **Page 3 (Investor Analytics):** Geographic transaction volume heatmaps and age group SIP splits.
- **Page 4 (SIP & Market Trends):** Dual-axis SIP inflow vs Nifty 50 benchmark overlay.

Figure 6.1: Dashboard Page 1 Screenshot - Industry Overview



7. Recommendations & Stakeholder Impact

1. **For Retail Investors:** Prioritize funds with Sortino Ratio > 1.2 and Calmar Ratio > 1.0 to ensure superior downside risk protection.
2. **For AMC Product Managers:** Monitor SIP continuity gaps > 35 days to trigger automated re-engagement workflows for at-risk investors.
3. **For Wealth Advisors:** Utilize the multi-factor fund recommendation engine (`scripts/recommender.py`) to align scheme selection with investor risk appetite.

8. Limitations

- Historical NAV history spans 4.5 years (2022-2026); longer 10-year market cycles could refine long-term CAGR estimates.
- Investor transaction data uses AMFI-anchored synthetic distributions for demographic modeling.